

STA322 - Project 1

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Introduction

Our project uses a two-stage probability proportional to size (PPS) cluster sample to estimate characteristics of faculty in Trinity College of Arts & Sciences at Duke University. We aimed to estimate four characteristics of this population: the fraction of faculty who are women, the fraction with at least one grant ongoing in 2026, the average number of years since receiving a Ph.D. among faculty who hold one, and the average number of years since last publication among faculty who have published. To answer these questions, we aimed for a representative sample of about 100 faculty, drawn in a way that limited how many departmental websites we had to visit.

Sampling design

Our goal was a final sample of 100 faculty, which we obtained in two stages. In the first stage, we selected 20 of the 41 Trinity departments using systematic PPS sampling without replacement, with each department's number of faculty as its measure of size. In the second stage, we selected 5 faculty from each chosen department by simple random sampling without replacement. This gives a final sample of $20 \times 5 = 100$ faculty.

```
library(survey)
library(tidyverse)
library(pps)
library(dplyr)
```

```
# load and reorganize data

trinity <- read_csv("trinity_faculty.csv")

trinity <- trinity |>
```

```

select(department = "Dept name", M_i = "# of faculty")

# certainty departments (pi_i = 1), same 7 as in the sampling step
certainty_depts <- c("Economics", "Physics", "Romance Studies", "Music",
                    "Biology", "Mathematics", "Psychology & Neuroscience")

certainty <- trinity |>
  filter(department %in% certainty_depts) |>
  mutate(pi_i = 1)

# remaining departments: 13 more selected, pi_i = 13 * M_i / sum(M_i)
remaining <- trinity |>
  filter(!department %in% certainty_depts)
remaining$pi_i <- 13 * remaining$M_i / sum(remaining$M_i)

dept_info <- bind_rows(certainty, remaining)

n_dept <- 20
t_x <- sum(trinity$M_i)
trinity$pi_i <- n_dept * trinity$M_i / t_x

max(trinity$pi_i) # should be 1.378, so some departments exceed 1

```

```
[1] 1.378151
```

Let M_i be the number of eligible faculty in department i , and let $M_0 = \sum_{i=1}^{41} M_i$ be the total number of eligible faculty across all 41 departments. If no department were large enough to require certainty selection, the first-order inclusion probability for department i under PPS sampling would be

$$\pi_i = \frac{20 M_i}{M_0}$$

For the largest departments, this formula gave $\pi_i > 1$, which is not a valid probability.

```

# dealing with certainty cases

# separate initial certainty and non-certainty departments
certainty <- trinity |>
  filter(pi_i >= 1) |>
  mutate(pi_i = 1)

remaining <- trinity |>

```

```

  filter(pi_i < 1)

# calculate number of additional departments to select
n_remaining <- n_dept - nrow(certainty)

# recalculate pi_i for remaining departments
remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)

# if recalculation creates additional certainty PSUs,
# repeat until all remaining probabilities are below 1
while (any(remaining$pi_i >= 1)) {
  new_certainty <- remaining |>
    filter(pi_i >= 1) |>
    mutate(pi_i = 1)
  certainty <- bind_rows(certainty, new_certainty)
  remaining <- remaining |>
    filter(pi_i < 1)
  n_remaining <- n_dept - nrow(certainty)
  remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)
}

```

We treated these departments as certainty PSUs and set $\pi_i = 1$ for them. We then recalculated the inclusion probabilities for the remaining departments, using the number of selections still needed and the total number of faculty in the remaining departments. This pushed one more department above 1, so we repeated the process until every remaining probability was below 1. In the end, 7 departments were certainty PSUs, leaving 13 departments to be selected.

```

# select remaining cases with pps & add certainty cases

set.seed(919)

# returns numbers of selected rows
selected_depts <- ppss(remaining$M_i, n_remaining)

# keep selected departments
pps_remaining <- remaining[selected_depts, ]

# combine certainty psus with pps-selected departments
pps_sample <- bind_rows(certainty, pps_remaining)

# check 20 departments were selected
nrow(pps_sample)

```

```
[1] 20
```

```
# view selected departments
pps_sample |>
  select(department, M_i, pi_i)
```

```
# A tibble: 20 x 3
  department                M_i pi_i
  <chr>                <dbl> <dbl>
1 Biology                77 1
2 Economics              81 1
3 Mathematics            82 1
4 Music                  62 1
5 Physics                60 1
6 Romance Studies        66 1
7 Psychology & Neuroscience 59 1
8 Art, Art History & Visual Studies 44 0.814
9 Asian & Middle Eastern Studies 45 0.832
10 Cinematic Arts         7 0.129
11 Computer Science       48 0.888
12 Dance Program          21 0.388
13 English               50 0.925
14 Gender, Sexuality & Feminist Studies 13 0.240
15 History               47 0.869
16 Philosophy             24 0.444
17 Political Science      43 0.795
18 Sociology              35 0.647
19 Statistical Science    32 0.592
20 Writing and Rhetoric Program 31 0.573
```

We set a random seed (919) before selecting the departments so that our sample can be reproduced. The 13 non-certainty departments selected this way, together with the 7 certainty departments, gave us our sample of 20 departments.

```
# stage 2 srs

# srs 5 faculty from each department
pps_sample$faculty_numbers <- lapply(
  pps_sample$M_i,
  function(M_i) sample(1:M_i, 5)
)
```

```
# view numbers of faculty selected
pps_sample |>
  select(department, M_i, pi_i, faculty_numbers)
```

```
# A tibble: 20 x 4
```

	department	M_i	pi_i	faculty_numbers
	<chr>	<dbl>	<dbl>	<list>
1	Biology	77	1	<int [5]>
2	Economics	81	1	<int [5]>
3	Mathematics	82	1	<int [5]>
4	Music	62	1	<int [5]>
5	Physics	60	1	<int [5]>
6	Romance Studies	66	1	<int [5]>
7	Psychology & Neuroscience	59	1	<int [5]>
8	Art, Art History & Visual Studies	44	0.814	<int [5]>
9	Asian & Middle Eastern Studies	45	0.832	<int [5]>
10	Cinematic Arts	7	0.129	<int [5]>
11	Computer Science	48	0.888	<int [5]>
12	Dance Program	21	0.388	<int [5]>
13	English	50	0.925	<int [5]>
14	Gender, Sexuality & Feminist Studies	13	0.240	<int [5]>
15	History	47	0.869	<int [5]>
16	Philosophy	24	0.444	<int [5]>
17	Political Science	43	0.795	<int [5]>
18	Sociology	35	0.647	<int [5]>
19	Statistical Science	32	0.592	<int [5]>
20	Writing and Rhetoric Program	31	0.573	<int [5]>

Within each selected department i , we selected 5 faculty by simple random sampling without replacement. The probability that faculty member j was chosen, given that department i was selected, was therefore

$$\pi_{j|i} = \frac{5}{M_i}.$$

The overall inclusion probability for faculty member j in department i is

$$\pi_{ij} = \pi_i \pi_{j|i} = \pi_i \cdot \frac{5}{M_i}.$$

Data Collection

```
#read and clean collected data

samples <- read.csv("pps_sample.csv")

samples <- samples |>
  rename(
    woman = Sex...1...F...0...other.,
    grant = Ongoing.grant...1...Yes..0...No.,
    has_phd = PhD...1...Yes..0...No.,
    yrs_phd = Number.of.years.since.PhD,
    yrs_pub = Years.since.last.pub,
    intl_ugrad = International.undergrad.degree.
  )

# attach M_i and pi_i, then build the weight
samples <- samples |>
  left_join(dept_info, by = "department")

samples$weight <- samples$M_i / (5 * samples$pi_i)

#dealing with missing data
# z = 1 if y is observed, 0 if missing; obs = y * z

# Q3: years since phd
samples$z_phd <- ifelse(!is.na(samples$yrs_phd)
  & samples$has_phd == 1, 1, 0)
samples$obs_phd <- ifelse(samples$z_phd == 1, samples$yrs_phd, 0)

# Q4a: years since last pub
samples$z_pub <- ifelse(!is.na(samples$yrs_pub), 1, 0)
samples$obs_pub <- ifelse(samples$z_pub == 1, samples$yrs_pub, 0)

fac_des <- svydesign(
  ids = ~department,
  weights = ~weight,
  data = samples
)
```

Let S be the set of sampled departments and s_i the set of sampled faculty in department i . For a variable y_{ij} , the estimated population total is

$$\hat{t}_y = \sum_{i \in S} \sum_{j \in s_i} w_{ij} y_{ij}$$

and the estimated population mean is

$$\hat{\bar{y}} = \frac{\sum_{i \in S} \sum_{j \in s_i} w_{ij} y_{ij}}{\sum_{i \in S} \sum_{j \in s_i} w_{ij}}$$

The fraction of faculty who are women, the fraction with an ongoing grant, and the fraction with an international undergraduate degree are weighted proportions. The average years since Ph.D. and the average years since last publication are weighted means, restricted to sampled faculty who hold a Ph.D. and to those who have published, respectively. We computed all estimates with the ‘survey’ package, using departments as the clusters, and report 95% confidence intervals.

Question 1. What is an estimate of the fraction of faculty who are women?

```
#|label: women-prop
```

```
prop_women <- svymean(~woman, fac_des)
prop_women
```

```
      mean      SE
woman 0.35968 0.062
```

```
confint(prop_women, level = .95)
```

```
      2.5 %    97.5 %
woman 0.2381951 0.4811584
```

We estimate that 36.0% of Trinity faculty are women. We are 95% confident that the true fraction of Trinity faculty who are women is between 24.8% and 48.1%.

Question 2. What fraction of faculty have at least one grant that is ongoing in 2026?

```
#|label: grant
```

```
prop_grant <- svymean(~grant, fac_des)
prop_grant
```

	mean	SE
grant	0.18953	0.0539

```
confint(prop_grant, level = .95)
```

	2.5 %	97.5 %
grant	0.08396691	0.2950893

We estimate that 19.0% of Trinity faculty have at least one grant ongoing in 2026. We are 95% confident that the true fraction of Trinity faculty with at least one grant ongoing in 2026 is between 8.4% and 29.5%.

Question 3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?

```
#|label: phd

avg_phd <- svyratio(~obs_phd, ~z_phd, fac_des)
avg_phd
```

Ratio estimator: svyratio.survey.design2(~obs_phd, ~z_phd, fac_des)

Ratios=

	z_phd
obs_phd	25.66529

SEs=

	z_phd
obs_phd	2.433472

```
confint(avg_phd, level = .95)
```

	2.5 %	97.5 %
obs_phd/z_phd	20.89577	30.43481

Among faculty who have a Ph.D., we estimate that the average time since receiving the degree is 25.7 years. We are 95% confident that, among Trinity faculty with a Ph.D., the true average number of years since receiving the degree is between 20.7 and 30.7 years.

Question 4

a) Among those who have published papers, what is the avg number of years since their last publication?

```
#|label: pub-yrs

avg_pub <- svyratio(~obs_pub, ~z_pub, fac_des)
avg_pub
```

```
Ratio estimator: svyratio.survey.design2(~obs_pub, ~z_pub, fac_des)
Ratios=
```

```
      z_pub
obs_pub 2.858112
SEs=
      z_pub
obs_pub 0.6312359
```

```
confint(avg_pub, level = .95)
```

```
      2.5 %    97.5 %
obs_pub/z_pub 1.620912 4.095311
```

Among faculty who have published papers, we estimate that the average time since their last publication is 2.8 years. We are 95% confident that, among Trinity faculty who have published papers, the true average number of years since their last publication is between 1.5 and 4.2 years.

b) What fraction of faculty received their undergraduate degree outside the United States?

```
#|label: intl-ug

prop_intl <- svymean(~intl_ugrad, fac_des, na.rm = TRUE)
prop_intl
```

```
      mean      SE
intl_ugrad 0.18539 0.0631
```

```
confint(prop_intl, level = .95)
```

```
      2.5 %    97.5 %
intl_ugrad 0.06174701 0.309031
```

We estimate that 18.5% of Trinity faculty with a known undergraduate institution received their undergraduate degree outside the United States. We are 95% confident that the true fraction is between 6.2% and 30.9%.

Caveats and other considerations

Systematic PPS sampling has a drawback for variance estimation. As with any systematic sample, some pairs of departments can never be selected together, so the joint inclusion probability π_{ik} is 0 for some pairs (i, k) . The usual Horvitz–Thompson variance estimator needs these joint probabilities, so it cannot be used to estimate the variance of our estimates. Following lecture, we instead pretend the sample was drawn with replacement and use the with-replacement variance estimator.

Initially, we considered a simple random sample (SRS) of 100 faculty. An SRS has equal inclusion probabilities and is easy to analyze. The problem is that it requires a complete list of every eligible Trinity faculty member before any sampling can happen, and the goal of our design was to avoid compiling faculty information from all 41 department websites, so an SRS did not fit our data collection.

We also considered stratified sampling, with each department as a stratum. Stratifying ensures every department is represented, and it can improve precision when faculty within a department are similar to each other but departments differ. However, department sizes vary a lot. With proportional allocation,

$$n_i = n \frac{M_i}{M_0}$$

many small departments would be allocated well under one faculty member. For example, Cinematic Arts, with 7 faculty, would get about 0.6. Rounding would either leave many small departments out or force us to move far from proportional allocation. Stratification would also still require a faculty list for all 41 departments.

We therefore chose a two-stage PPS cluster design, which balances statistical representation against practical effort. Sampling departments with probability proportional to size gives larger departments a better chance of being selected, and the second-stage SRS gives a probability sample of faculty within each selected department. Most importantly, we only need to build faculty lists for the 20 selected departments instead of all 41. Choosing 20 departments, rather than fewer, also spreads the sample over many clusters, which limits the loss of precision that happens when faculty within the same department are similar.